

AI FASHION DESIGN GENERATOR

AI-powered fashion creation, recommendation and virtual try-on
IBM Internship 2026 | Developed by Piyush Ramteke

INTRODUCTION & BACKGROUND

Fashion design ideation is time-consuming
Creating and iterating on design concepts manually takes significant time and expertise, slowing down the creative process for designers at all levels.

Traditional visualization tools require specialist skills
Existing design software demands technical proficiency, creating barriers for non-expert users and limiting accessibility to professional designers only.

Generative AI can unify fragmented workflows
Product discovery, virtual try-on, and design generation exist as separate tools today. Generative AI offers the opportunity to combine these stages into a single, seamless platform.

PROBLEM STATEMENT

1. Traditional Workflow:
2. Idea → Manual Design → Visualization → Product Search → Try-On
3. Key Problems:
4. • Time-consuming design process
5. • Fragmented, disconnected tools
6. • Difficult visualization without specialist skills
7. • Limited accessibility for non-experts
8. • Separate product discovery and try-on steps

PROPOSED SOLUTION

User Fashion Idea

- ↓ Gemini Fashion Specification Extraction
- ↓ AI Fashion Image Generation

- ↓ Product Recommendation
- ↓ AI Virtual Try-On

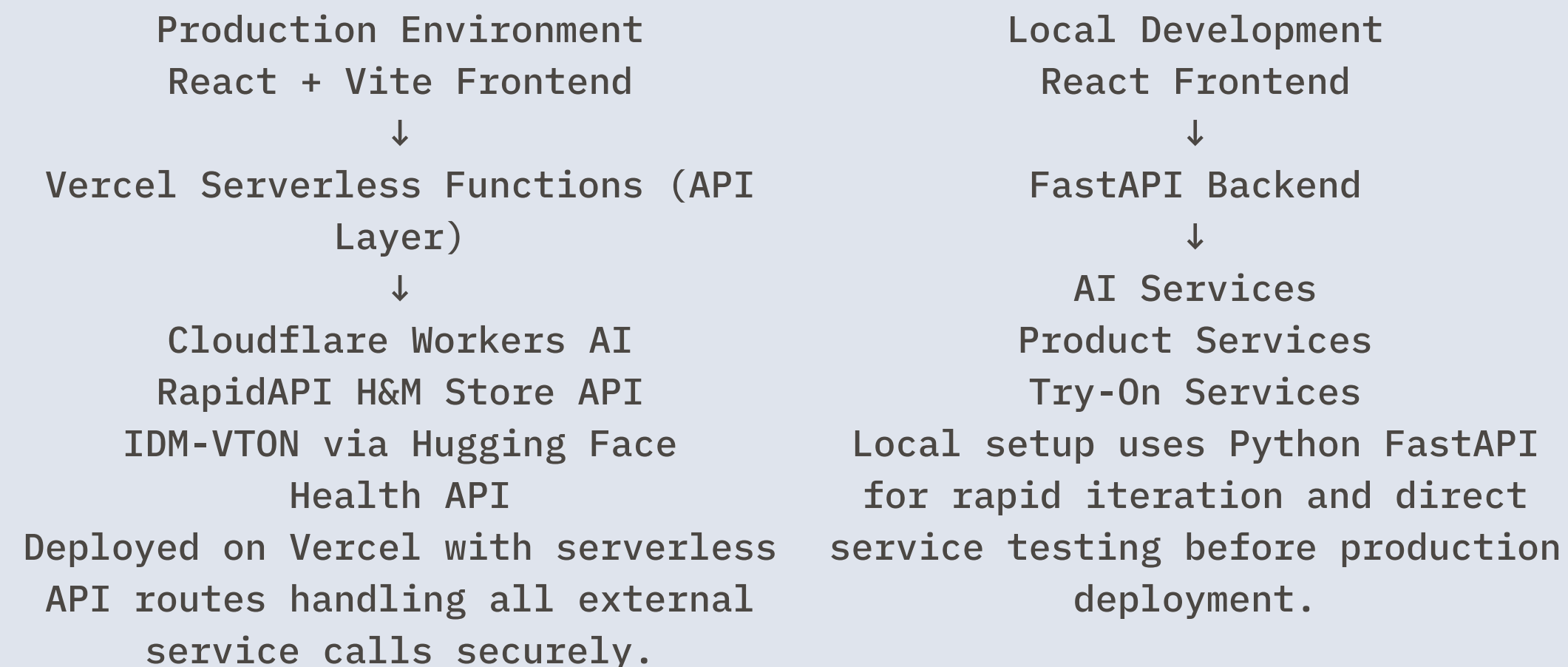
↓ Collection / Tech Pack Generation

A unified end-to-end pipeline powered by Generative AI

PROJECT OBJECTIVES

- Convert natural-language fashion descriptions into structured specifications
- Generate fashion design concepts using Generative AI models
- Retrieve and rank matching fashion products via RapidAPI H&M Store API
- Provide AI-assisted virtual try-on functionality using IDM-VTON
- Enable users to save designs and generate technical information (tech packs)
- Deliver a unified end-to-end fashion ideation workflow

SYSTEM ARCHITECTURE



TECHNOLOGY STACK

Frontend

- React 19
- Vite 8
- Tailwind CSS 4
- Lucide React

Backend

- Python & FastAPI
- Pydantic & HTTPX
- SlowAPI

AI Models

- Google Gemini 2.5

Flash

- Cloudflare Workers AI
- IDM-VTON

External Services

- Hugging Face Spaces
- RapidAPI (H&M Store)
- Vercel (Deployment)

AI FASHION DESIGN PIPELINE

Input Stage

- Natural Language Prompt – User describes fashion idea in plain text
- Google Gemini 2.5 Flash – Extracts structured fashion attributes (style, color, garment type, fit, fabric)
- Fashion Specification – Structured JSON output ready for downstream processing

Generation Stage

- Prompt Optimization – Specification is converted into an optimized image generation prompt
- Cloudflare Workers AI – Hosts image generation models (FLUX.1, SDXL, DreamShaper)
- Generated Fashion Image – High-quality AI-rendered fashion design output

Key Insight

- Gemini extracts structured attributes from free-form text, ensuring consistent specification quality
- Cloudflare Workers AI generates images serverlessly, enabling scalable production deployment
- The two-stage pipeline decouples understanding from generation for modularity and accuracy

AI MODEL SELECTION

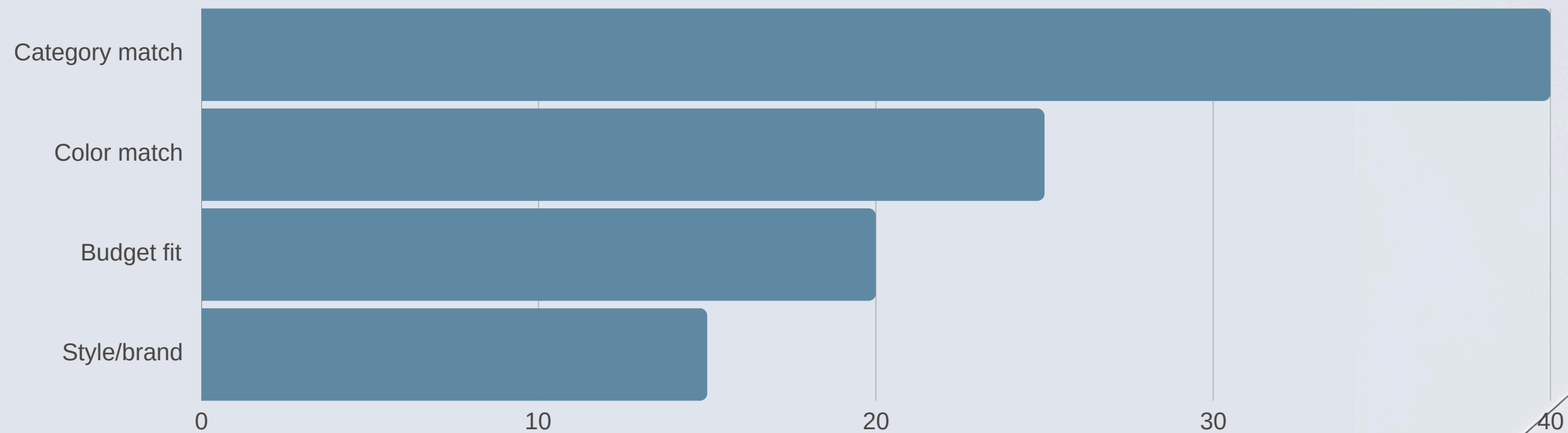
FLUX.1 Schnell: Fast image synthesis optimized for rapid fashion concept generation with minimal latency.

Stable Diffusion XL Base 1.0: General-purpose high-quality image generation for detailed and realistic fashion designs.

DreamShaper 8 LCM: Creative style generation enabling expressive and artistic fashion design concepts.

Stable Diffusion XL Lightning: Enhanced detail and speed for high-fidelity fashion image output with improved rendering performance.

PRODUCT RECOMMENDATION WORKFLOW



Source: AI-generated data. Replace or verify before use.

Workflow: Fashion Specification → RapidAPI H&M Store API → Product Candidates → Filtering & Ranking → Top Recommendations
Ranking uses structured metadata only – no vector embeddings.
Signals are weighted to prioritize category and color alignment, ensuring highly relevant product matches for the user's fashion specification.

VIRTUAL TRY-ON INTEGRATION

Workflow: Person Photo + Garment Image → IDM-VTON via Hugging Face → AI Try-On Result

The user provides a person photo and a garment image; IDM-VTON processes both and returns a photorealistic AI-generated try-on result.

Integration Details:

- Third-party IDM-VTON model accessed via `gradio_client`
- Auto-cropping applied to input images before processing
- Server-side processing with ZeroGPU queue and retry logic
- Handles cold starts through robust error and queue handling

License and Constraints:

- IDM-VTON licensed under CC BY-NC-SA 4.0 (non-commercial use)
- Works best with clean, garment-only images
- AI-generated scene images may produce variable try-on results
- Subject to Hugging Face Spaces availability and ZeroGPU quotas

USER WORKFLOW & APPLICATION SCREENSHOTS

Dashboard UI

- Central hub displaying recent designs, saved collections, and quick-access tools
- Wardrobe Quick Picks and AI Studio entry points prominently featured

Runway View

- Full-screen showcase of AI-generated fashion designs
- Side-by-side comparison of multiple generated concepts

Studio Workspace

- Natural language prompt input for fashion description
- Model selector (FLUX.1, SD XL, DreamShaper 8 LCM, SD XL Lightning)
 - Real-time fashion specification extraction display

Product Recommendation Interface

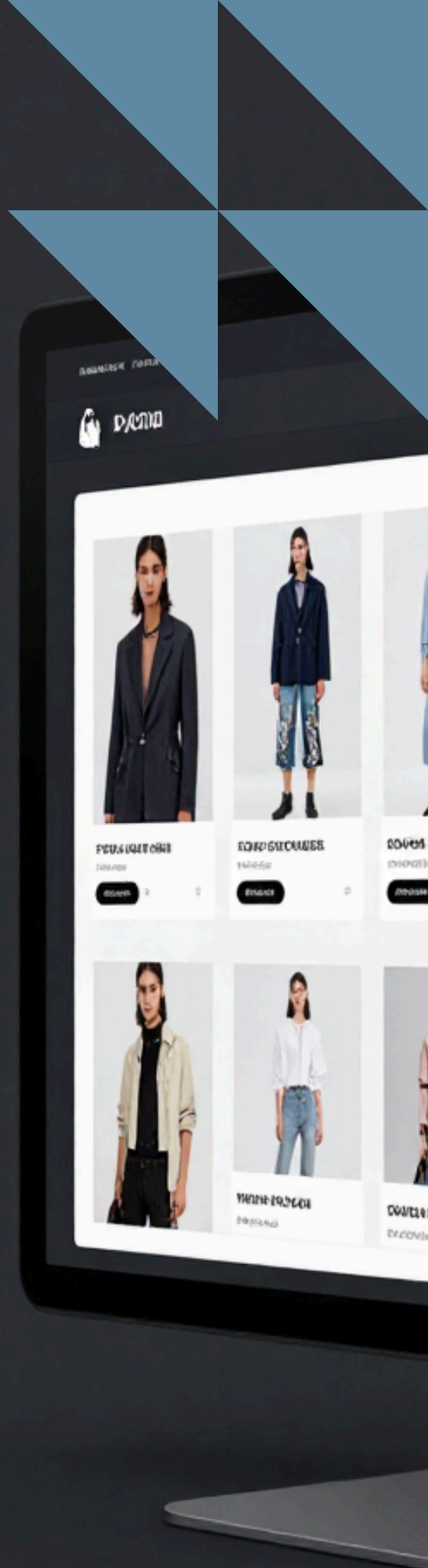
- Grid layout of matched H&M products with category, color, and budget filters
 - Ranked results with match score indicators

Virtual Try-On Results

- Before/After comparison view of person photo and garment
 - IDM-VTON output with download and copy options

Saved Collections View

- Gallery of saved designs with tech pack generation and export options



KEY FEATURES

- AI Design Studio for fashion creation
- Fashion Specification Extraction from text
- Multiple AI Image Generation Models
- AI Product Recommendations via H&M API
- Virtual Try-On using IDM-VTON
- Before and After Try-On Comparison
- Save Collections and Download Designs
- Copy Fashion Specification Text
- Tech Pack Generation
- Wardrobe Quick Picks
- Security Controls and Validation
- Responsive Web Interface

SECURITY & RELIABILITY

API Credentials & Data Protection

- Server-side API credentials management – keys never exposed to client
- Environment variables used for all secrets and sensitive configuration
- File-type validation enforced on all user uploads

Access Control & Rate Limiting

- 10 MB limit enforced on image uploads to prevent abuse
- Rate limiting applied to all API endpoints via SlowAPI
- CORS restrictions configured to allow only trusted origins

Error Handling & Reliability

- Structured error handling with safe, non-leaking API error responses
- ZeroGPU retry logic and queue handling for IDM-VTON try-on stability
- No committed node_modules or secrets in repository

TESTING AND VALIDATION

- Backend unit tests
 - Mocked API endpoint tests
 - Build verification scripts
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- API input validation
 - Error handling tests
 - End-to-end workflow verification

CHALLENGES & SOLUTIONS

1.IDM-VTON API Integration → Correct Gradio parameter mapping and gradio_client usage resolved compatibility issues with the Hugging Face endpoint.

2.ZeroGPU Cold Starts → Implemented retry logic and queue/error handling to manage GPU availability delays gracefully.

3.Product Recommendation → RapidAPI H&M integration with structured ranking logic (category, color, budget, style) enabled reliable results.

4.Frontend/Backend Deployment → Vercel serverless API for production plus FastAPI local testing ensured consistent behavior across environments.

DEVELOPMENT EVOLUTION

Early Stages

Initial Prototype → Cloudflare AI Integration → UI/UX

Redesign

Foundational setup, first AI image generation, and interface overhaul to improve usability and visual design.

Core Integration

Real IDM-VTON Integration → Vercel Serverless API Deployment
→ RapidAPI Product Recommendations

Replaced mocks with live try-on model; deployed serverless API layer; added real H&M product recommendation pipeline.

Finalization

Security & Documentation Improvements → Final QA and UX

Enhancements

Hardened API credential handling, improved validation, refined user experience, and completed project documentation.

CURRENT RESULTS

- End-to-end AI fashion ideation and visualization workflow
- AI-generated fashion images via Cloudflare Workers AI
- Real fashion product API integration with H&M (RapidAPI)
- AI virtual try-on via IDM-VTON on Hugging Face
- Responsive React web interface with Tailwind CSS
- Download and copy utilities for designs and specs
- Production-oriented Vercel serverless API architecture
- Security controls, validation, and deployment documentation

LIMITATIONS

IDM-VTON works best with clean garment-only images; AI-generated scene images may produce variable or inconsistent virtual try-on results.

Third-party APIs (H&M, Hugging Face, Gemini) have usage quotas and availability limits that may affect reliability.

Product recommendations depend on H&M API catalog scope and availability. Gemini extraction depends on configured API access and prompt quality.

This project is an internship prototype and is not intended as a commercial production platform.

FUTURE SCOPE

Improved garment extraction and preprocessing for cleaner VTON inputs

Enhanced VTON preprocessing pipeline for higher-quality try-on results

Integration with more fashion marketplaces beyond H&M
Personalized product recommendations based on user style history

User authentication and cloud-based saved collections

Advanced analytics dashboard for design and usage insights

Mobile application development for on-the-go fashion ideation

Expanded AI model support for diverse design aesthetics

Collaborative features for team-based fashion projects

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THANK
YOU

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